

CONCEPT NOTE

Agentic AI Smart Weather & Pest Alert System for Farmers

Aligned SDGs: SDG 2 (Zero Hunger)
SDG 9 (Industry, Innovation and Infrastructure)
SDG 12 (Responsible Consumption & Production)
SDG 13 (Climate Action),
SDG 15 (Life on Land)

Category: Agentic AI System

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Project Title: Agentic AI Smart Weather & Pest Alert System for Farmers

1. Problem Statement

Agriculture is highly dependent on weather conditions. Farmers often suffer significant crop losses due to unexpected rainfall, heatwaves, storms, droughts, and pest outbreaks. In many rural areas, farmers do not receive weather forecasts or pest alerts in time to take preventive measures.

Traditional methods of receiving agricultural information rely on television, radio, newspapers, or manual announcements by agricultural officers. These methods are often delayed, unavailable in local languages, or fail to provide location-specific recommendations.

There is a need for an intelligent, automated, and real-time alert system that continuously monitors weather conditions, predicts risks, and instantly notifies farmers in their local language with actionable recommendations.

2. Proposed Solution

The **Agentic AI Smart Weather & Pest Alert System** is an automated solution that provides farmers with real-time weather forecasts and timely pest alerts using **n8n workflow automation** and the **OpenWeather API**. The system automatically collects weather data such as temperature, rainfall, humidity, and wind speed at regular intervals. It then analyzes the data using predefined rules and identifies weather conditions that may affect crop growth, such as heavy rain, heatwaves, or high humidity.

When a potential risk is detected, the system generates personalized recommendations for farmers, including irrigation scheduling, pesticide spraying, and crop protection measures. These alerts are instantly delivered through **SMS or WhatsApp** in the farmer's preferred local language, ensuring that important information reaches them quickly and is easy to understand.

Since the entire process is automated using **n8n**, no manual intervention is required. The system continuously monitors weather conditions, sends real-time notifications, and helps farmers make informed decisions to reduce crop losses, improve productivity, and adopt sustainable farming practices. This solution can also be used by agricultural organizations and government agencies to provide location-specific advisories to a large number of farmers efficiently.

3. Key Features

- Real-time weather monitoring
- Automatic OpenWeather API integration
- JSON-based workflow automation
- Heavy rain detection
- Heatwave detection
- Humidity monitoring
- Wind speed monitoring
- Pest risk recommendations
- Irrigation scheduling
- Spraying recommendations
- SMS alerts
- Local language support
- Fully automated workflow
- Low-cost implementation
- Scalable for multiple villages

4. Learning Roadmap Agent

- Collect historical weather and farming data.
- Analyze weather patterns and crop conditions.
- Learn from farmer feedback to improve recommendations.
- Provide crop-specific farming guidance.
- Recommend irrigation and spraying schedules.
- Predict seasonal weather risks.

- Generate personalized farming advisories.
- Continuously improve decision-making using AI.
- Help farmers adopt best agricultural practices.
- Update recommendations based on new weather forecasts.

5. Technology Stack

➤ Frontend

- React.js
- HTML5
- CSS3
- JavaScript

➤ Backend

- Node.js
- Express.js
- n8n Workflow Automation
- JavaScript

➤ Database

- MongoDB
- Google Sheets

➤ Artificial Intelligence

- Rule-Based Decision Engine
- OpenAI GPT
- Google Gemini

➤ APIs & Services

- OpenWeather API
- WhatsApp Business API
- Twilio SMS API
- Google Maps API

6. Benefits

- Provides real-time weather and pest alerts.
- Reduces crop damage due to adverse weather.
- Improves irrigation planning.
- Optimizes pesticide and fertilizer usage.
- Saves time through automated alerts.
- Reduces farming costs.

- Increases crop productivity and yield.
- Supports local-language notifications.
- Promotes sustainable farming practices.
- Helps agricultural organizations communicate efficiently with farmers.

7. Expected Impact

The proposed **Agentic AI Smart Weather & Pest Alert System** will help farmers make better farming decisions by providing timely weather forecasts and pest alerts. Early notifications about heavy rain, heatwaves, or unfavorable weather conditions will enable farmers to take preventive measures, reducing crop damage and improving overall productivity.

The system will also promote efficient use of water, fertilizers, and pesticides by recommending the right time for irrigation and spraying. This will lower farming costs, reduce resource wastage, and encourage sustainable agricultural practices while protecting the environment.

Additionally, agricultural organizations and government agencies can use the platform to deliver location-specific advisories to a large number of farmers. Overall, the system will improve farmers' income, increase climate resilience, strengthen food security, and support the adoption of smart farming technologies.

8. Target Users

➤ Primary Users

- Farmers
- Small-scale Farmers
- Large-scale Farmers

➤ Secondary Users

- Agriculture Organizations
- Government Agriculture Departments
- NGOs
- Farmer Producer Organizations (FPOs)
- Agricultural Extension Officers
- Cooperatives

9. Future Scope

- Integrate AI and Machine Learning for accurate pest and disease prediction.
- Connect IoT sensors to monitor soil moisture, temperature, and humidity.
- Use satellite imagery and drones for crop health monitoring.
- Develop a mobile application for farmers.
- Provide voice-based alerts in regional languages.

- Add crop yield prediction and fertilizer recommendations.
- Integrate GPS-based location-specific weather alerts.
- Include an AI chatbot for 24/7 farming assistance.
- Support offline SMS alerts for areas with poor internet connectivity.
- Expand the system to support multiple crops and regions.

10. Conclusion

The **Agentic AI Smart Weather & Pest Alert System** provides an intelligent, automated solution for delivering timely weather and pest advisories to farmers. By combining **n8n workflow automation**, **JSON-based logic**, and the **OpenWeather API**, the system continuously monitors environmental conditions, analyzes weather risks, and sends localized alerts through SMS or WhatsApp. This enables farmers to make informed decisions about irrigation, pesticide spraying, and crop protection, reducing losses and improving productivity. The solution is scalable, cost-effective, and supports climate-smart agriculture, making it suitable for deployment by individual farmers, agricultural organizations, and government agencies.

